



Fig. 4: Illustrations of electrical designs created with the help of VariCAD

electrical symbols, welding and tolerance symbols, etc.

3D-2D export. To produce conventional drafting documentation, 3D models can be easily converted into 2D drawings. By defining the views in 3D, 2D views of one or more selected solids can be created. In addition, exporting specified sections is possible. After changes in 3D, VariCAD supports updates of a 2D drawing.

2D drawing and editing. In VariCAD, the drawing functions are optimised for ease of use in engineering. Some handy features of 2D drawing include tolerance symbols, welding

symbols, finish symbols, advanced dimensioning, ortho mode, hatching with automatic border detection, drawing layers, block creation, rectangular grid, auxiliary construction lines, automatic detection of objects and snap points and numerous snap modes. Drawings can be made in millimetres or inches.

BOM and title blocks. To maintain the data structure of the product, VariCAD provides certain tools. There are links between attributes of parts and content of title blocks. The user can create a BOM from an assembly or easily modify the database using commands like sort-

ing of information and mass attribute changes. Each part can contain attributes, like name, supplier or type of material. Such data can be used for filling of title blocks, creation of BOM, material requisitions or other purposes.

It is possible to export the data structure of the product (BOM) into other systems or into a spreadsheet. A mask is used for BOM customisation where a user can modify it exactly according to the requirements.

Compatibility

VariCAD can interchange files with other CAD systems. One can export STEP (3D), STL (3D), IGES (3D), DWG (2D) and DXF (2D) files, and import STEP (3D), DWG (2D) and DXF (2D) files. The files can either be converted individually or in batch routines

(converting multiple files in one step).

VariCAD is available in English, Japanese, Chinese, Portuguese and German versions. It uses DXF, DWG (AutoCAD), IGES, STL and STEP file formats for data exchange. The files exported from VariCAD can be further processed in rendering, FEA or CAM systems.

VariCAD can be run on MS Windows or Linux (both 32- and 64-bit versions). It uses Unicode, which supports other fonts (Japanese, Chinese, Cyrillic, etc) and special characters used in technical drawings. ●

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